

```
Q1a) if (grade >= 90) {  
System.out.println("Great Job!");  
}
```

```
b) if (num < 20 || num > 50 ){  
System.out.println("Error");  
}
```

```
c) if (y < 100) {  
y = y + 2;  
System.out.print ("y=" + y);  
}
```

```
Q2 if (num1 > num2) {  
System.out.println("First number is larger.");  
} else if (num1 < num2) {  
System.out.println("Second number is larger.");  
} else {  
System.out.println("Numbers are equal.")  
}
```

Q3 a) even odd

```
b) if (num % 2 != 0) {  
System.out.println("odd number");  
} else {  
System.out.println("even number");  
}
```

```
Q4 a) int num = rand.nextInt(50) + 1;  
b) int num = rand.nextInt(81) + 20;  
c) double num = rand.nextDouble() * 10 + 10;
```

Q5 ages 18 and 65 are not covered

```
If (age <= 18) {  
    System.out.println("child");  
} Else if (age < 65) {  
    System.out.println("adult");  
} Else if (age >= 65) {  
    System.out.println("senior");  
}
```

Q6 a) T

b) F

c) T

- d) T
- e) T
- f) T
- g) T

Q8

- a) T
- b) F because it's between two integers, roundoff errors would not occur among integers.
- c) F because if statement is an interdependent logic, thus if-else if statement is a mutually exclusive
- d) F because switch only can be applied to byte, short, int, char, string, enum.
- e) T
- f) F because the same seed produces the same result each time.
- g) T
- h) T
- i) F because the ! is evaluated before &&
- j) T
- k) T
- l) T